

to be rapidly built, changed, enhanced and deployed.

Open source cloud makes enterprise data available openly, so that external parties can build services with adequate visibility of the enterprise's functioning and availability of interfaces.

Based on the various customer engagements I have had in the cloud migration space, I would recommend the following:

- Choose the right cloud provider; each provider has specific strengths and weaknesses. Evaluate these properly so that they meet customer needs.
- Use the right migration tools.
- No vendor lock-in of the application.
- Move to an agile operating model.
- Replace data centres of customers with private and hybrid clouds.
- Maximise productivity with scalability and high availability.

- Always migrate development and testing environments to the cloud.
- Move core functions to the private cloud and non-core to the public cloud.
- Wherever possible, adopt a hybrid model – it's a safe bet.
- Transform architecture to a scale out capability, enabled for dynamic API

based interaction.

- Bring cloud native capabilities to the application.
- Reduce costs by using a higher level of automation.
- Have skilled staff that knows all the new technologies and processes, e.g., microservices, DevOps, API management, AI, etc. **END** 

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One of the biggest challenges facing blockchain technology is scalability. As the users and transactions on a blockchain network grow, the network can become slower and less efficient. Research is underway to develop new scaling solutions, such as sharding and layer-two scaling, which could help blockchain networks handle larger volumes of transactions.

Interoperability is another challenge. There are many different blockchain networks and protocols, and they do not always work well together. Standards and protocols that can enable blockchain networks to interoperate with each other, allowing for more seamless transfer of data and value between networks, are being researched.

Security and privacy are also critical concerns for blockchain technology. Researchers are working on developing new cryptographic techniques and protocols that can enhance the security and privacy of blockchain networks. This

Table 1: Additional platforms for blockchain development

Platform	URL
Kaleido	https://www.kaleido.io/
Venly	https://www.venly.io/
Blockdaemon	https://blockdaemon.com/
Xooa	https://xooa.com/
Alchemy	https://www.alchemy.com/

includes developing new methods for authentication, encryption, and consensus algorithms. DeFi (decentralised finance) is a fast-growing sector of the blockchain industry that is focused on developing financial applications that run on decentralised blockchain networks. New DeFi protocols and applications are being developed, such as decentralised exchanges, lending platforms, and stablecoins.

As blockchain networks become more complex and decentralised, the question of governance becomes

increasingly important. Research is underway to develop new governance models and frameworks that can ensure the security, stability, and fairness of blockchain networks.

The scope for research and development in blockchain is broad and multi-disciplinary, encompassing fields such as computer science, cryptography, economics, law, and more. As blockchain technology continues to evolve and mature, it is likely that new areas of research and development will emerge, while existing areas continue to be refined and expanded. **END** 

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The author is associated with various academic and research institutes for delivering expert lectures and conducting technical workshops on the latest technologies and tools.